

Credit Score Classification Report

➤ Project Overview

In This project builds end-to-end machine learning classification model that predicts a customers credit score based on their financial and behavioral profile. The system covers the full pipeline, data cleaning and exploratory data analysis, feature engineering, training and comparing multiple classification algorithms, selecting the best-performing model , and deploying it behind a web application with a REST API and an interactive HTML front end.

The Target variable has 3 classes :

- Good – the customer has a healthy credit profile
- Standard – the customer has an average credit profile
- Poor – the customer has a weak credit profile

➤ Dataset and Preprocessing

There are 2 dataset are given for this work , 1st one is Train.csv and the 2nd one is Test.csv . This 2 dataset each contains around 27-28 columns combining numeric and categorical data type. Identifier columns such as ID, Customer_ID, Month , SSN , and Name were dropped since they carry no predictive value.

❖ **Dataset And Preprocessing**

2 datasets were used, a training dataset and a test dataset. Each originally containing 27-28 columns combining numeric and categorical data types.

❖ **Missing Value Handling**

In this project , basically I used to handling all the missing values for both dataset. Using mean , standard deviation , mode , median for numeric datatype , and bfill, ffill, and median for string columns alone side handling placeholder value like -100-- ,--20 etc .

❖ **Encoding**

All categorical columns were converted to numeric from using scikit-learn's LabelEncoding. Each encoder was fit on the combined train and test values for that columns so that both datasets share a consistent, alphabetically ordered mapping. The target column, Credit Score was separately label-encoded into 0 = good, 1 = poor and 2= standard.

❖ **Train\Test split**

After cleaning and encoding , the training dataset was split into a training set and a hold-out test set using a 75% /25% split(random forest = 42) for model evaluation.

➤ Feature Used

The final model was trained on the following 21 input features

Age	Customer's age (cleaned of stray underscores, converted to numeric)
Occupation	Customer's profession (label-encoded)
Annual_Income	Reported annual income
Monthly_Inhand_Salary	Monthly take-home salary (missing values filled with mean)
Num_Bank_Accounts	Number of bank accounts held
Num_Credit_Card	Number of credit cards held
Interest_Rate	Interest rate on existing credit
Num_of_Loan	Number of loans currently held
Type_of_Loan	Type(s) of loan (missing values filled with mode)
Delay_from_due_date	Average delay in days from the payment due date
Num_of_Delayed_Payment	Number of delayed payments
Changed_Credit_Limit	Recent change in credit limit
Num_Credit_Inquiries	Number of credit inquiries made
Credit_Mix	Quality of the credit mix (Bad / Standard / Good)
Outstanding_Debt	Total outstanding debt
Credit_Utilization_Ratio	Ratio of credit used to credit available
Credit_History_Age	Length of credit history, converted from text (e.g. "3 Years and 6 Months") to total months
Payment_of_Min_Amount	Whether only the minimum payment is made (Yes / No / NM)
Total_EMI_per_month	Total EMI obligations per month
Amount_invested_monthly	Monthly investment amount
Payment_Behaviour	Spending and payment behaviour category
Monthly_Balance	Remaining monthly balance

➤ Modelling Approach

4 classification algorithms were trained and compared on the same train/test split in order to select the best-performing model:

- ✓ **Decision Tree Classifier** : Entropy criterion, max depth 40, used as a baseline model.
- ✓ **Random Forest Classifier**: 500 trees, max depth 20, with bootstrap sampling.
- ✓ **Random Forest tuned with GridSearchCV**: hyper-parameter search over number of estimator and tree depth, using 5-fold cross-validation.
- ✓ **XGBoost Classifier**: 1000 estimators, learning rate 0.03, max depth 6, multi-class object.
- ✓ **CatBoost Classifier**: 20000 estimators, learning rate 0.05, depth 10 with early stopping: this was ultimately selected as the final model.

For every model, performance was measured with accuracy score, a full classification report (precision, recall, F1-score per class), and a confusion matrix plotted as a heatmap.

➤ Results

Model	Training Accuracy	Test Accuracy	Notes
Decision Tree	75.45%	69.02%	Baseline model, entropy criterion, max depth 40
Random Forest	86.98%	75.85%	500 trees, max depth 20
Random Forest + GridSearchCV	86.98%	75.85%	Hyper-parameter tuned, same result as base RF
XGBoost	83.12%	75.43%	1000 estimators, learning rate 0.03
CatBoost (Final / Best Model)	–	77.52%	2000 iterations, depth 10, selected for deployment

CatBoost achieved the highest test accuracy (77.52%) among all models evaluated and was therefore chosen as the production model. It was serialized with Python's pickle module and saved as `best_model.pkl` for deployment.

➤ Deployment Architecture

The Trained Catboost model is deployed as a lightweight web service build with Flask, paired with a browser-based front end

❖ Backend (app.py)

- Loads `best_model.pkl` once at start-up.
- Re-implements the same categorical encodings used during training (Occupation, Type_of_Loan, Credit_Mix, Payment_of_Min_Amount, Payment_Behaviour), sorted alphabetically to mirror LabelEncoder's behaviour.
- Converts the Credit_History_Age text format back into total months using the same regular expression logic as the training notebook.
- Exposes three routes:
 - GET / – serves the index.html front end
 - GET /health – returns a simple status check
 - POST /predict – accepts a JSON payload of customer features and returns the predicted credit score with class probabilities

The `/predict` endpoint returns the predicted label (0, 1, or 2), the corresponding credit score name (Good / Poor / Standard), the model's confidence for the predicted class, and the full probability distribution across all three classes.

❖ Frontend (index.html)

A single HTML page provides a form for entering a customer's financial details and displays the prediction results returned by the /predict API, including the predicted credit score and confidence level.

❖ Project Files

- final_project_2.ipynb – the Jupyter notebook containing data cleaning, exploratory analysis, model training, evaluation, and model export.
- best_model.pkl – the final trained CatBoost model, serialized with pickle.
- app.py – the Flask backend that loads the model and serves predictions.
- index.html – the web front end used to submit customer data and view predictions.

1. Conclusion

This project demonstrates a complete supervised machine learning workflow for credit score classification, from raw, messy financial data through cleaning, encoding, model comparison, and finally deployment as a working web application. Among the four algorithms tested, CatBoost provided the best generalization performance and was selected for production use.

Output

❖ InterFace

The screenshot displays the web interface of the Agrani Bank PLC Credit Score Prediction System. The interface is divided into several sections:

- Header:** Includes the Agrani Bank PLC logo, the text "Credit Intelligence & Assessment System — Powered by CatBoost ML", and navigation links for Home, About, Methodology, Reports, Documentation, Help, and Contact.
- Left Sidebar:** Contains a "QUICK LINKS" section with links to Credit Assessment, Score History, Reports, Methodology, Feature Guide, API Docs, and User Manual. Below this is an "INFORMATION" section with links to Score Guide, FAQ, and Contact Us.
- Main Content Area:**
 - PERSONAL & FINANCIAL INFORMATION:** A form with fields for AGE (28), OCCUPATION (Scientist), ANNUAL INCOME (\$) (8000), MONTHLY INHAND SALARY (\$) (4500), NUM BANK ACCOUNTS (3), NUM CREDIT CARDS (2), INTEREST RATE (%) (14), NUM OF LOANS (2), TYPE OF LOAN (Auto Loan), and a dropdown for Auto Loan.
 - PAYMENT & CREDIT HISTORY:** A form with fields for DELAY FROM DUE DATE (DAYS) (5), NUM DELAYED PAYMENTS (2), CHANGED CREDIT LIMIT (0.5), NUM CREDIT INQUIRIES (1), CREDIT MIX (Good), OUTSTANDING DEBT (\$) (1200), CREDIT UTILIZATION RATIO (%) (28.5), CREDIT HISTORY AGE (3 Years and 6 Months), PAYMENT OF MIN AMOUNT (Yes), TOTAL EMV / MONTH (\$) (450), AMOUNT INVESTED MONTHLY (\$) (200), and PAYMENT BEHAVIOUR (High_Limit_Direct_Payments).
 - CREDIT ASSESSMENT RESULT:** A section showing a "STANDARD" score with a confidence of 80.0%. A gauge chart displays the score distribution: GOOD (16.2%), STANDARD (80.0%), and POOR (3.8%).
 - SCORE LEGEND:** A table mapping scores to meanings: Good (Low risk borrower), Standard (Moderate risk), and Poor (High risk borrower).
- Footer:** Includes the Agrani Bank PLC logo, the text "Credit Intelligence & Assessment System — Powered by CatBoost ML", and navigation links for Home, About, Methodology, Reports, Documentation, Help, and Contact.

ⓘ Please fill in all required fields accurately. This system uses CatBoost ML to predict credit score as Good, Standard, or Poor.

PERSONAL & FINANCIAL INFORMATION

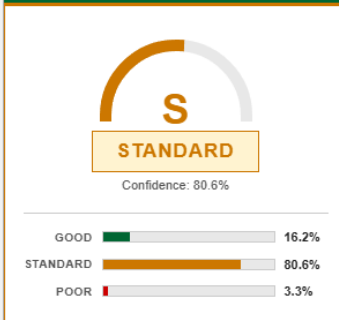
AGE	OCCUPATION
28	Scientist
ANNUAL INCOME (\$)	MONTHLY INHAND SALARY (\$)
60000	4500
NUM BANK ACCOUNTS	NUM CREDIT CARDS
3	2
INTEREST RATE (%)	NUM OF LOANS
14	2
TYPE OF LOAN	
Auto Loan	

PAYMENT & CREDIT HISTORY

DELAY FROM DUE DATE (DAYS)	NUM DELAYED PAYMENTS
5	2
CHANGED CREDIT LIMIT	NUM CREDIT INQUIRIES
0.5	1
CREDIT MIX	OUTSTANDING DEBT (\$)
Good	1200
CREDIT UTILIZATION RATIO (%)	CREDIT HISTORY AGE
28.5	3 Years and 6 Months
PAYMENT OF MIN AMOUNT	TOTAL EMI / MONTH (\$)
Yes	450
AMOUNT INVESTED MONTHLY (\$)	PAYMENT BEHAVIOUR
200	High_spent_Small_value_payments
MONTHLY BALANCE (\$)	
300	

🔍 ANALYZE CREDIT SCORE

CREDIT ASSESSMENT RESULT



● API online · <http://localhost:5000/predict>

SCORE LEGEND

Score	Meaning
Good	Low risk borrower
Standard	Moderate risk
Poor	High risk borrower

❖ Confusion Matrix

